

Big Bend Seagrasses Aquatic Preserve

Seagrass and Water Quality Trend Analysis

Florida SEACAR

2026-08-11

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Funding & Acknowledgements

The data used in this analysis is from the Export Standardized Tables in the SEACAR Data Discovery Interface (DDI). Documents and information available through the SEACAR DDI are owned by the data provider(s) and users are expected to provide appropriate credit following accepted citation formats. Users are encouraged to access data to maximize utilization of gained knowledge, reducing redundant research and facilitating partnerships and scientific innovation.

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Watershed Designations

Water quality sampling locations were divided into watershed designations based on recommendations from Big Bend Seagrasses Aquatic Preserve (BBSAP) staff, with influence from *Seagrass Integrated Mapping and Monitoring Program Mapping and Monitoring Report No. 2*¹. Submerged Aquatic Vegetation sampling locations contain watershed-based prefixes within their location names and naturally fall within these boundaries. BBSAP staff considered salinity, tidal exchange, and seasonal variation when designating sample locations as river or estuary.

SEACAR Data Methods

Data Matching

Data accessed through the Florida Department of Environmental Protection (FDEP) Watershed Information Network (WIN) are matched to a specific SEACAR program, when possible, based on a combination of multiple fields (e.g., WIN OrgID, ProjectID, SamplingAgency, MonitoringLocationID, etc.). WIN datasets that are not matched to a specific program page in the SEACAR DDI are maintained in the generic *Florida STORET/WIN*² program.

Data De-duplication Process

Coordination and overlap between monitoring programs in data collection activities has led to cases where datasets from two or more programs/data sources include some of the same result records (i.e., data from the same sampling event), leading to some risk of data duplication in the SEACAR data export tables. To mitigate this risk, a de-duplication process is run on the final export tables to check for and identify duplicate records (i.e., those with the same location, date, parameter, activity depth, species, quad/site identifier, and result values) before the tables are made public.

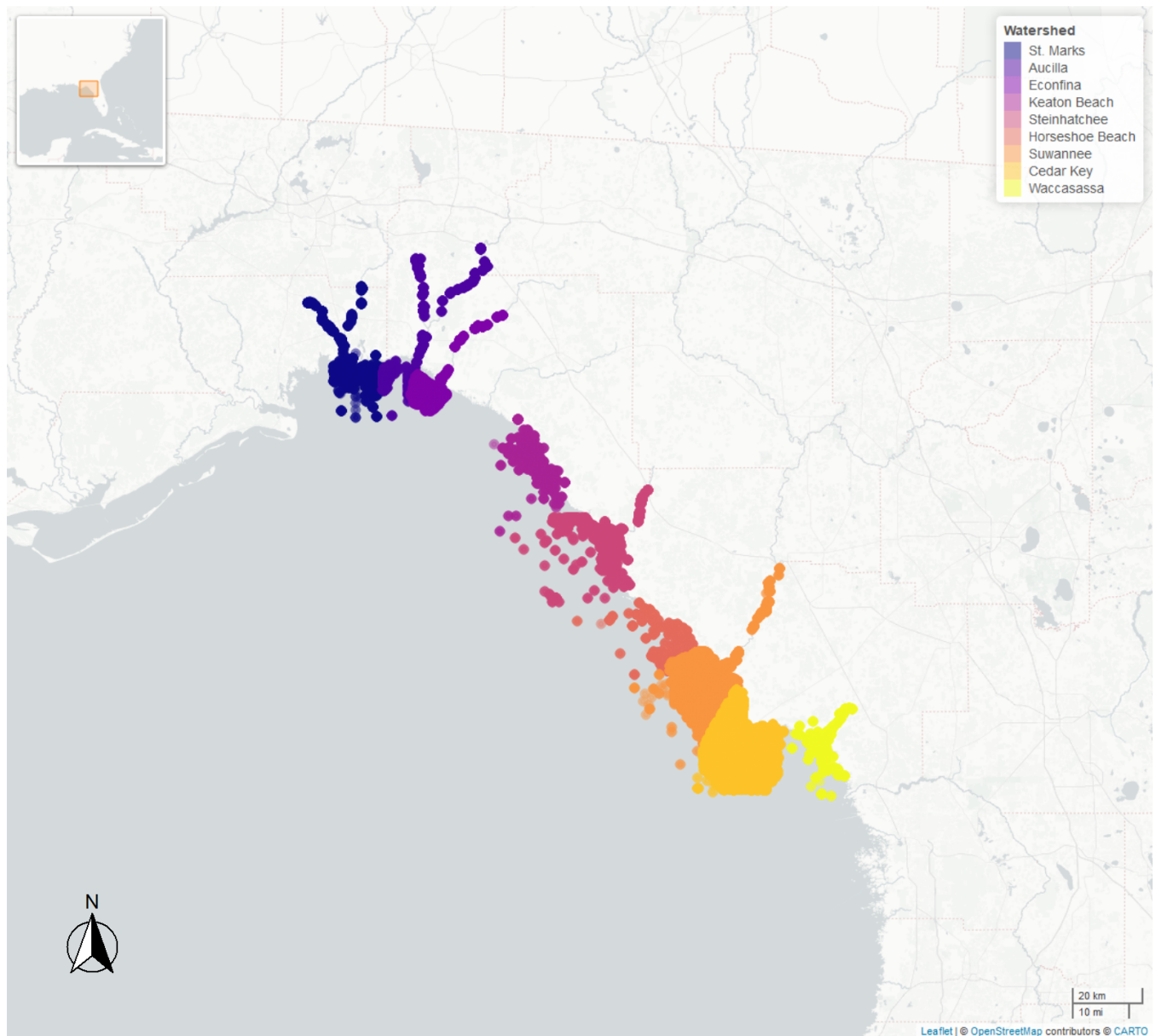
Seasonal Kendall-Tau Analysis

For discrete Water Quality data, indicators must have a minimum of ten years of data within the geographic range of the analysis to be included in the analysis. If there are insufficient years of data, the number of years of data

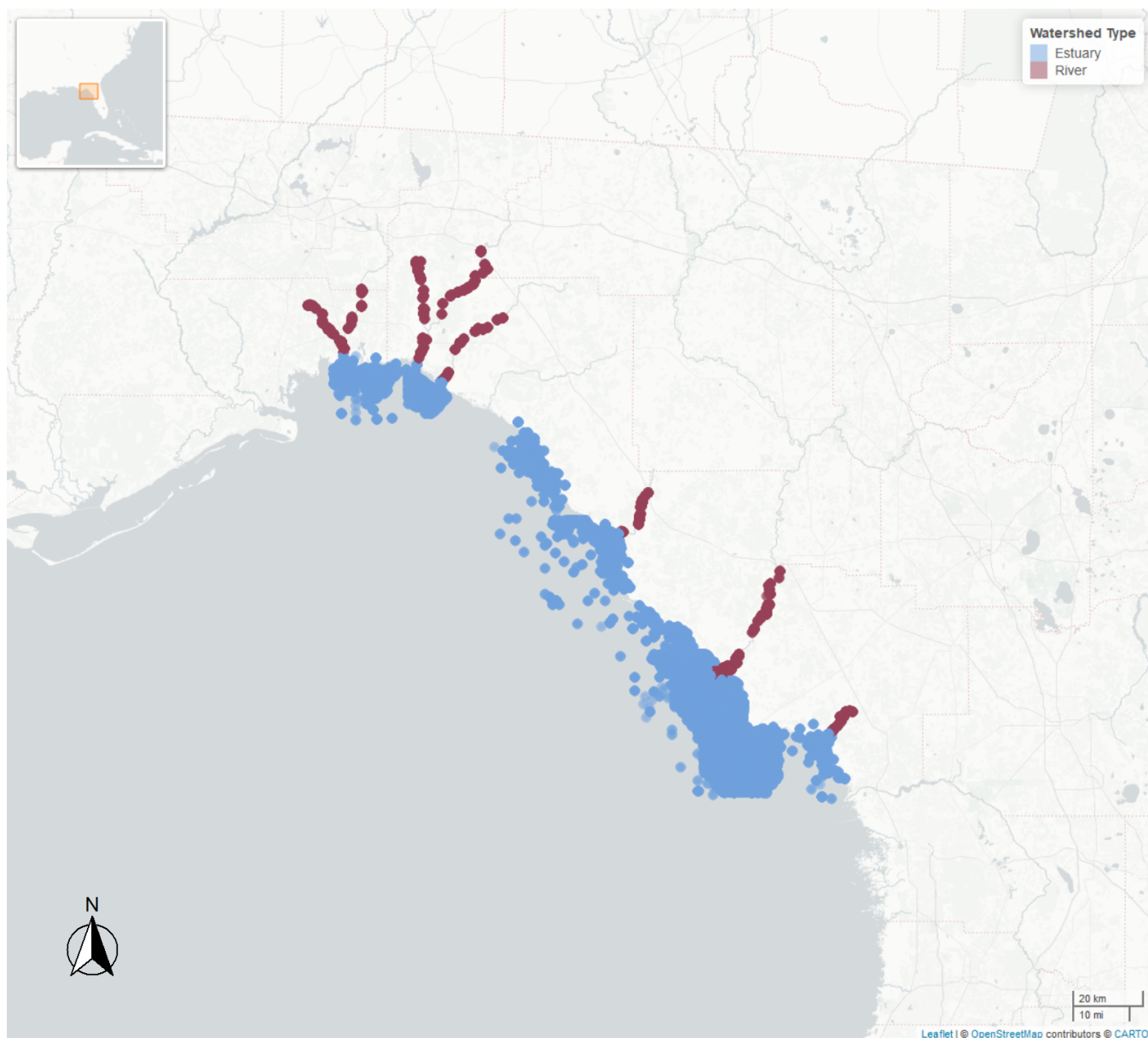
available will be noted and labeled as “insufficient data to conduct analysis”. Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Quality plots displayed in this report are monthly averages. Trend significance will be denoted as “Significant Trend” (when $p < 0.05$), or “Non-significant Trend” (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Maps

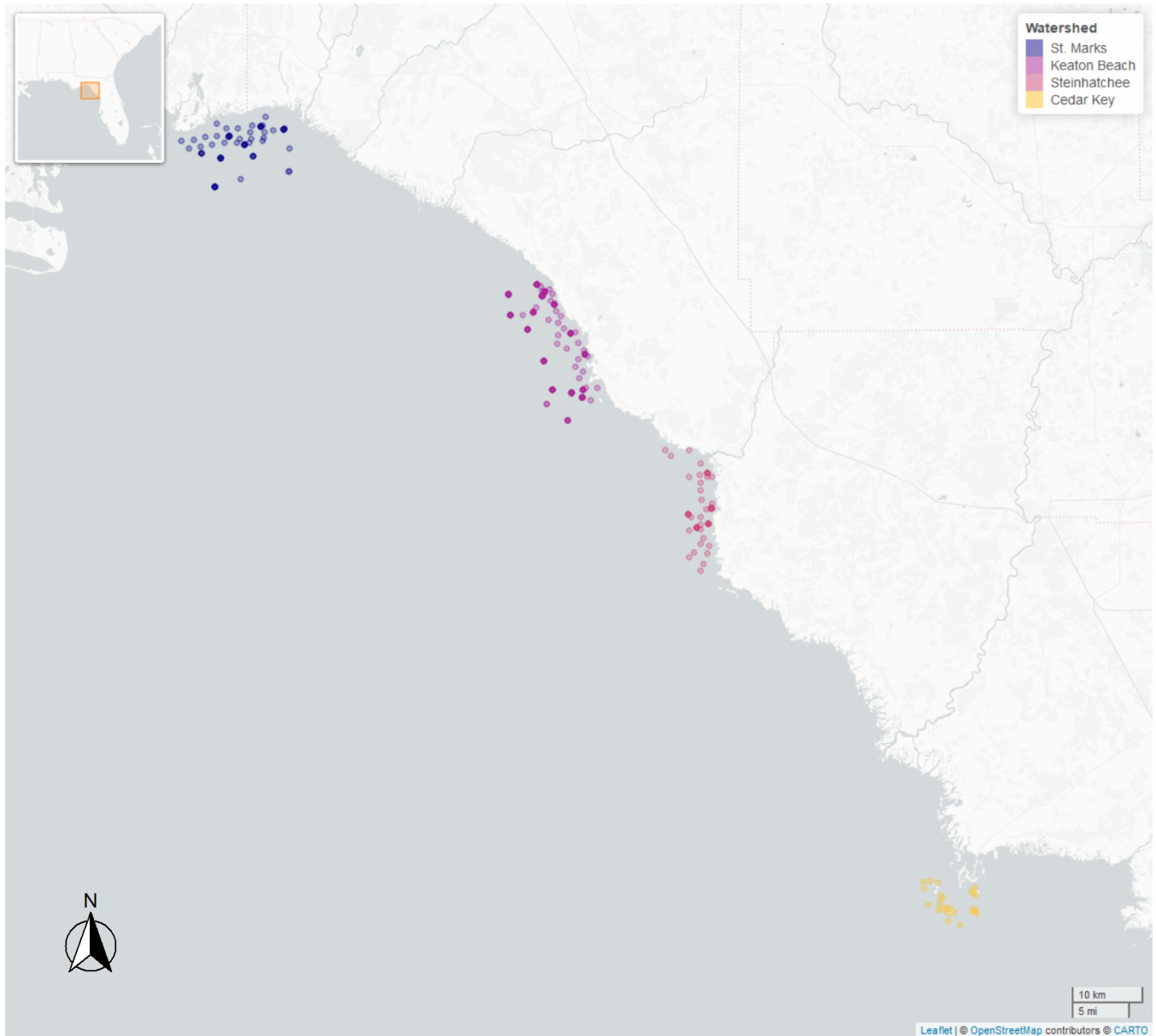
Water Quality Discrete - Sample Locations - All Parameters



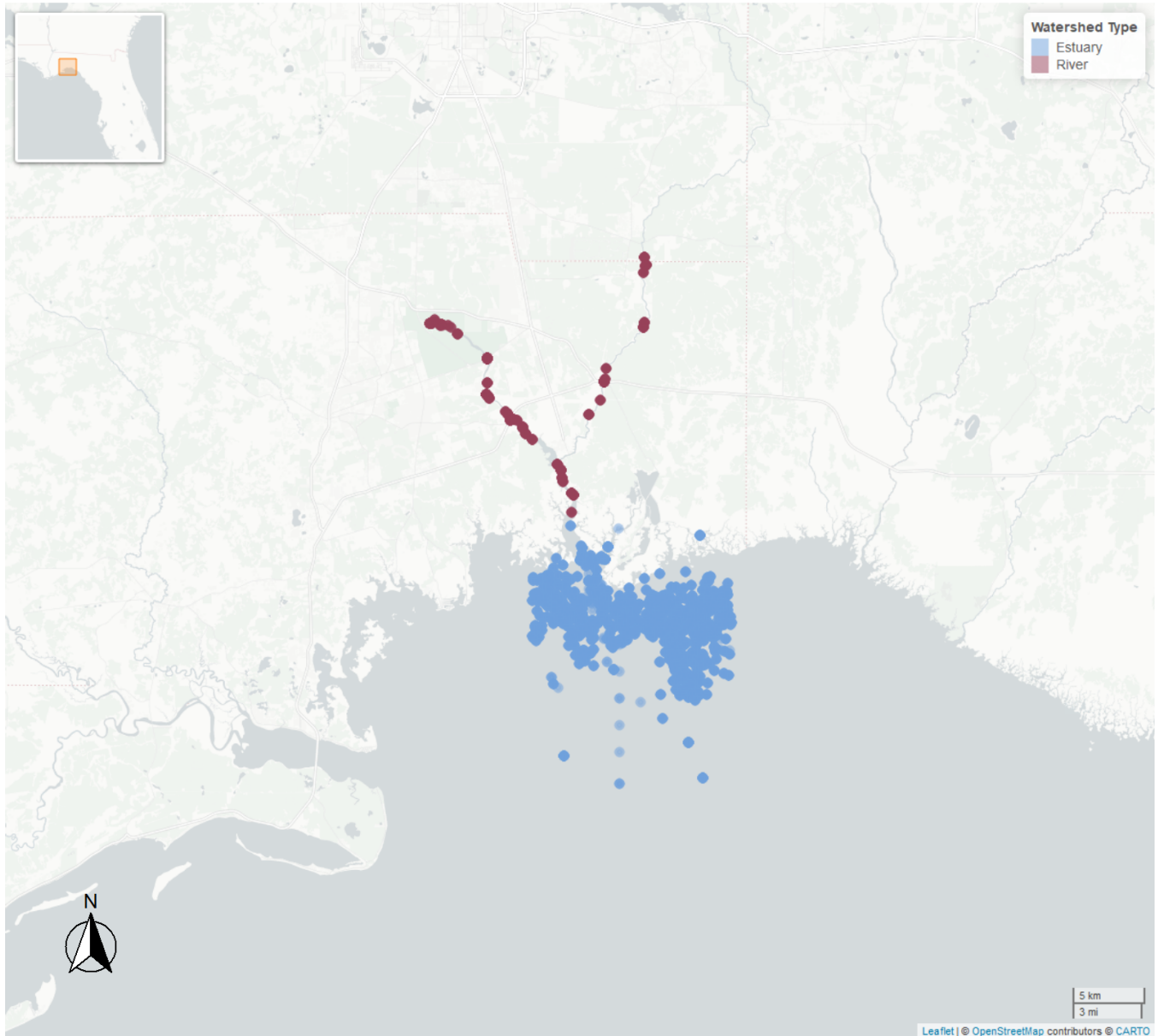
Water Quality Discrete - Sample Locations - Estuary and River Delineations



Submerged Aquatic Vegetation - Sample Locations



St. Marks



Plots will be produced below for all parameters. A Seasonal Kendall-Tau trendline will be plotted only for parameters where *SufficientData* is **TRUE**.

Trend arrows denote trends where the p value is less than 0.05 and the slope is less than 10% of the median parameter value.

St. Marks contains a *large trend*. Trends marked in $\uparrow$ or $\downarrow$ are deemed *large trends*, meaning the p value is less than 0.05 and the slope is greater than 10% of the median value.

Table 1: Seasonal Kendall-Tau Results for St. Marks

Type	ParameterName	Period-of-Record	N-Years	N-Data	SufficientData	SennSlope	SennIntercept	p	Trend
Estuary	Chlorophyll a, Corrected for Pheophytin	2005 - 2025	10	119	TRUE	0.17	1.36	0.0009	↑
River	Chlorophyll a, Corrected for Pheophytin	1999 - 2025	27	1312	TRUE	-0.01	0.92	0.0015	↓
Estuary	Chlorophyll a, Uncorrected for Pheophytin	2001 - 2025	12	138	TRUE	0.13	3.51	0.1204	0
River	Chlorophyll a, Uncorrected for Pheophytin	1996 - 2025	30	2603	TRUE	0.01	0.59	0.0007	↑
Estuary	Colored Dissolved Organic Matter	2001 - 2025	5	28	FALSE	-	-	-	-
River	Colored Dissolved Organic Matter	2001 - 2025	25	2073	TRUE	0.52	3.36	0.0000	↑
Estuary	Dissolved Oxygen	1993 - 2025	21	2430	TRUE	0.00	6.22	0.8856	0
River	Dissolved Oxygen	2001 - 2025	24	1380	TRUE	-0.11	6.52	0.0000	↓
Estuary	Dissolved Oxygen Saturation	2001 - 2025	9	111	FALSE	-	-	-	-
River	Dissolved Oxygen Saturation	2001 - 2025	15	575	TRUE	-2.31	97.68	0.0000	↓
Estuary	Salinity	1964 - 2025	28	2852	TRUE	-0.15	28.52	0.0052	↓
River	Salinity	2000 - 2025	21	886	TRUE	-0.06	3.87	0.0007	↓
Estuary	Secchi Depth	1993 - 2025	19	2071	TRUE	-0.01	2.15	0.1980	0
River	Secchi Depth	1996 - 2025	30	2551	TRUE	-0.02	4.21	0.0015	↓
Estuary	Total Nitrogen	2001 - 2025	13	166	TRUE	0.01	0.39	0.4935	0
River	Total Nitrogen	1992 - 2025	31	4251	TRUE	0.00	0.30	0.3657	0
Estuary	Total Phosphorus	2001 - 2025	13	157	TRUE	0.00	0.02	0.0559	0
River	Total Phosphorus	1992 - 2025	31	2733	TRUE	0.00	0.00	0.0000	↑
Estuary	Total Suspended Solids	2005 - 2025	10	120	TRUE	-0.38	10.63	0.0032	↓
River	Total Suspended Solids	1992 - 2025	29	831	TRUE	-0.08	5.48	0.0000	↓
Estuary	Turbidity	2005 - 2025	13	478	TRUE	0.03	1.72	0.1719	0
River	Turbidity	1992 - 2025	29	846	TRUE	0.00	0.75	0.0976	0
Estuary	Water Temperature	1964 - 2025	23	2770	TRUE	0.03	20.51	0.1974	0
River	Water Temperature	2001 - 2025	24	1378	TRUE	0.01	20.73	0.4680	0
Estuary	pH	1964 - 2025	21	2248	TRUE	0.00	7.95	0.5639	0
River	pH	2001 - 2025	24	1153	TRUE	0.00	7.57	0.2615	0

Table 2: Overview of Program Data for St. Marks

ParameterName	ProgramID	n-data-Estuary	n-data-River
Ammonium (NH4)	103	5	-
Chlorophyll a, Corrected for Pheophytin	514	-	237
Chlorophyll a, Corrected for Pheophytin	5002	119	1076
Chlorophyll a, Uncorrected for Pheophytin	103	8	1
Chlorophyll a, Uncorrected for Pheophytin	115	1	-
Chlorophyll a, Uncorrected for Pheophytin	118	5	-
Chlorophyll a, Uncorrected for Pheophytin	514	28	1959
Chlorophyll a, Uncorrected for Pheophytin	5002	98	644
Colored Dissolved Organic Matter	514	8	797
Colored Dissolved Organic Matter	5002	20	1277
Dissolved Oxygen	69	1988	-
Dissolved Oxygen	95	4	-
Dissolved Oxygen	103	2	3
Dissolved Oxygen	115	5	-
Dissolved Oxygen	118	16	3
Dissolved Oxygen	560	204	-
Dissolved Oxygen	5002	428	1539
Dissolved Oxygen Saturation	5002	111	575
NO2+3, Filtered	103	1	1
NO2+3, Filtered	5002	118	1863
Nitrate (NO3)	5002	-	3
Nitrite (NO2)	5002	-	10
Nitrogen, inorganic	103	6	1
Phosphate, Filtered (PO4)	5002	44	296
Salinity	69	1995	-
Salinity	95	21	-
Salinity	115	5	-
Salinity	118	15	1
Salinity	560	206	-
Salinity	5002	610	885
Secchi Depth	69	1952	-
Secchi Depth	103	2	-
Secchi Depth	115	3	-
Secchi Depth	118	2	-
Secchi Depth	514	28	1946
Secchi Depth	560	25	-
Secchi Depth	5002	86	700
Specific Conductivity	69	1986	-
Specific Conductivity	514	8	97
Specific Conductivity	5002	138	3183

Total Ammonia (N)	103	1	1
Total Ammonia (N)	5002	115	744
Total Kjeldahl Nitrogen	5002	115	764
Total Nitrogen	103	16	1
Total Nitrogen	118	2	-
Total Nitrogen	514	28	1977
Total Nitrogen	5002	122	2274
Total Phosphorus	103	13	1
Total Phosphorus	514	28	1959
Total Phosphorus	5002	116	774
Total Suspended Solids	5002	120	831
Turbidity	103	6	-
Turbidity	5002	472	851
Water Temperature	69	1995	-
Water Temperature	95	139	-
Water Temperature	115	5	-
Water Temperature	118	13	5
Water Temperature	560	205	-
Water Temperature	5002	631	1542
pH	69	1986	-
pH	95	1	-
pH	103	2	4
pH	115	5	-
pH	118	8	3
pH	560	167	-
pH	5002	255	1304

Program names:

69 - Fisheries-Independent Monitoring (FIM) Program³

95 - Harmful Algal Bloom Marine Observation Network⁴

103 - EPA STOrage and RETrieval Data Warehouse (STORET)/WQX⁵

115 - Environmental Monitoring Assessment Program⁶

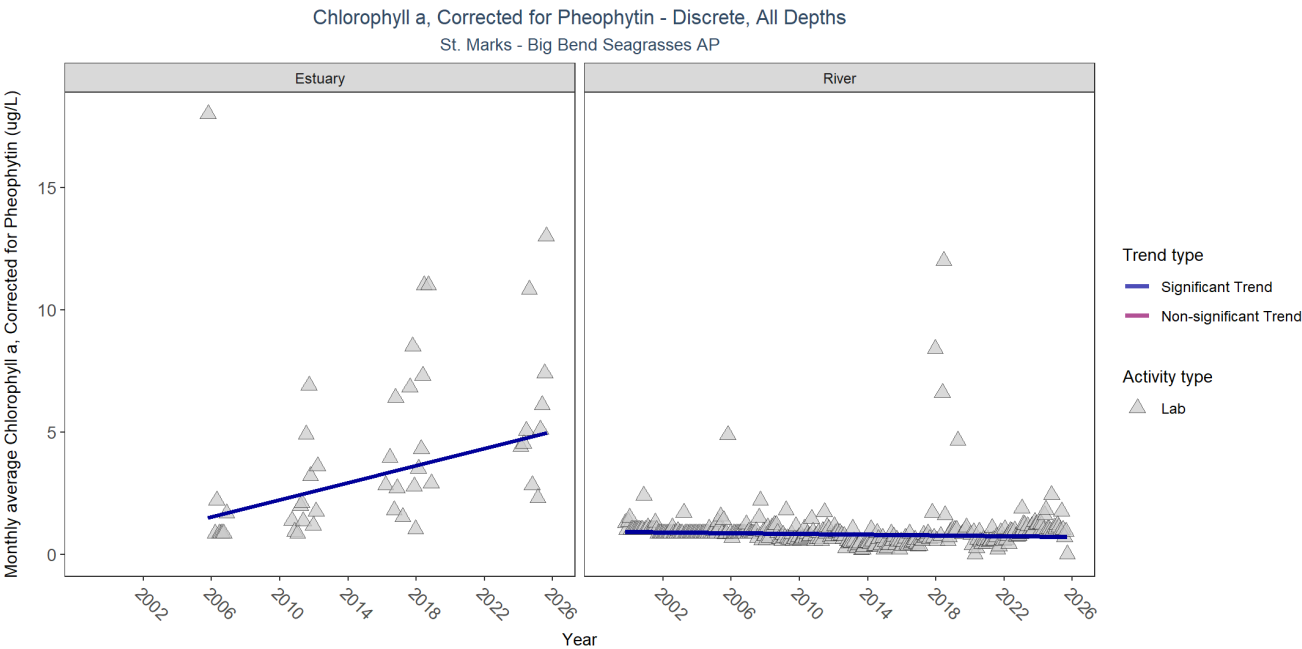
118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁷

514 - Florida LAKEWATCH Program⁸

560 - Big Bend Seagrasses & Nature Coast Aquatic Preserves - Seagrass Monitoring⁹

5002 - Florida STORET / WIN²

Chlorophyll a, Corrected for Pheophytin

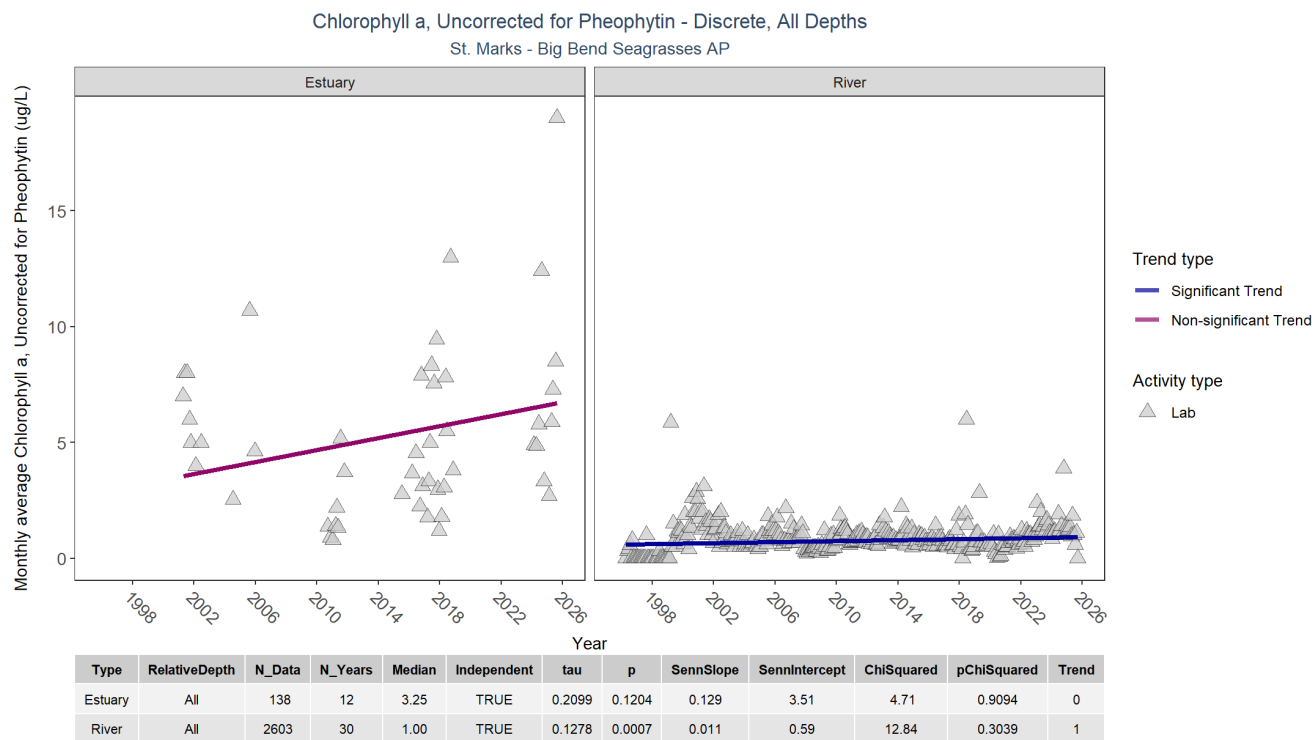


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	119	10	2.50	TRUE	0.4667	0.0009	0.175	1.36	12.78	0.2361	1
River	All	1312	27	0.84	TRUE	-0.1309	0.0015	-0.008	0.92	3.51	0.9822	-1

$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

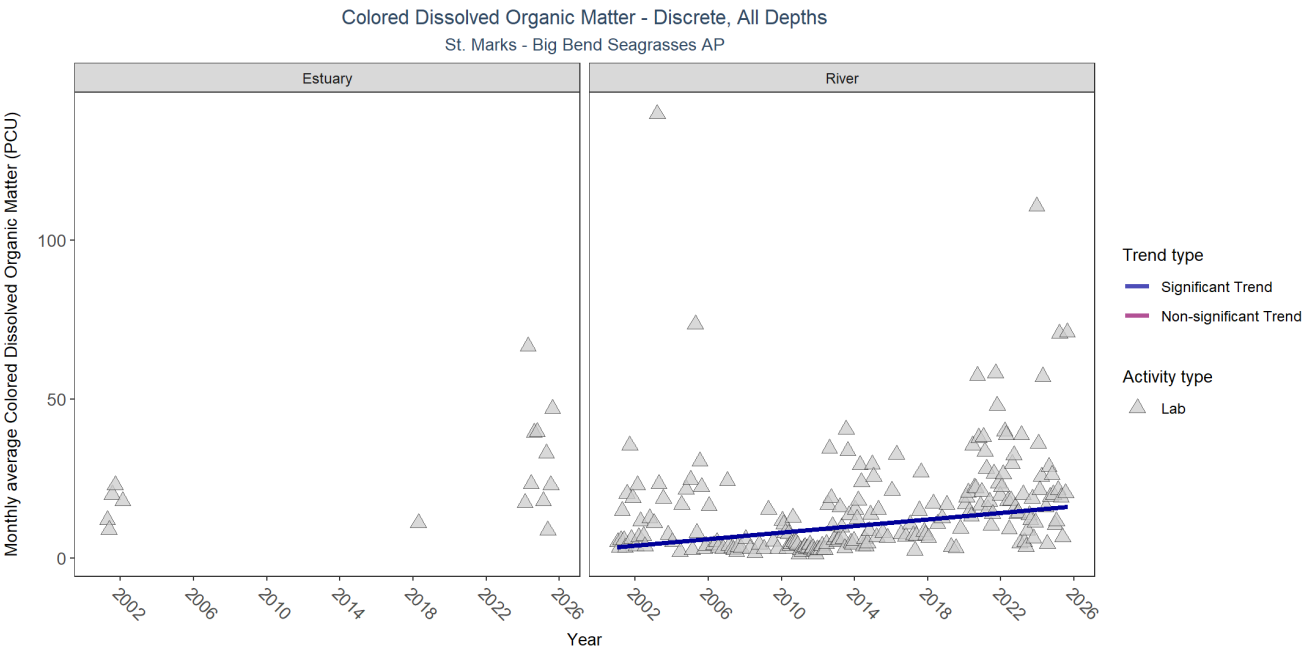
Chlorophyll a, Uncorrected for Pheophytin



$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

Colored Dissolved Organic Matter

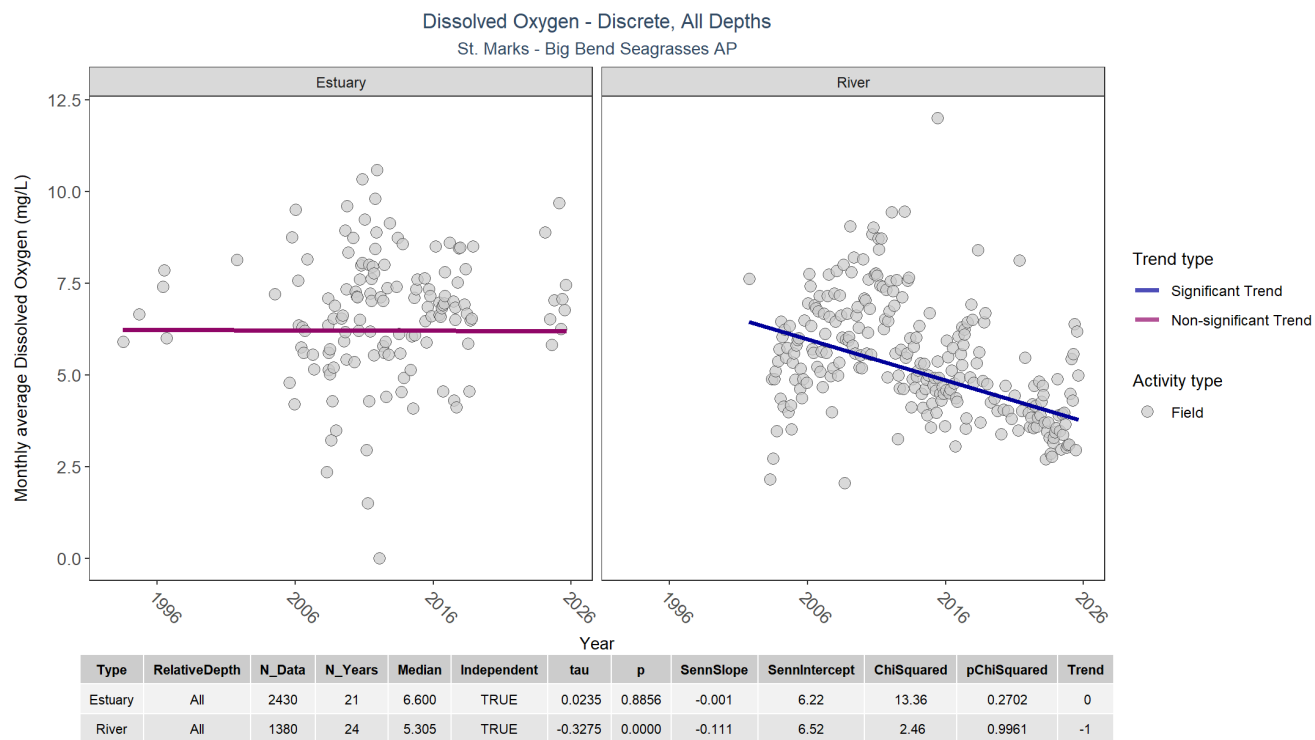


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	28	5	21.5	NA	NA	NA	NA	NA	NA	NA	NA
River	All	2073	25	15.0	TRUE	0.3061	0.0000	0.52	3.36	5.04	0.9294	1

$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

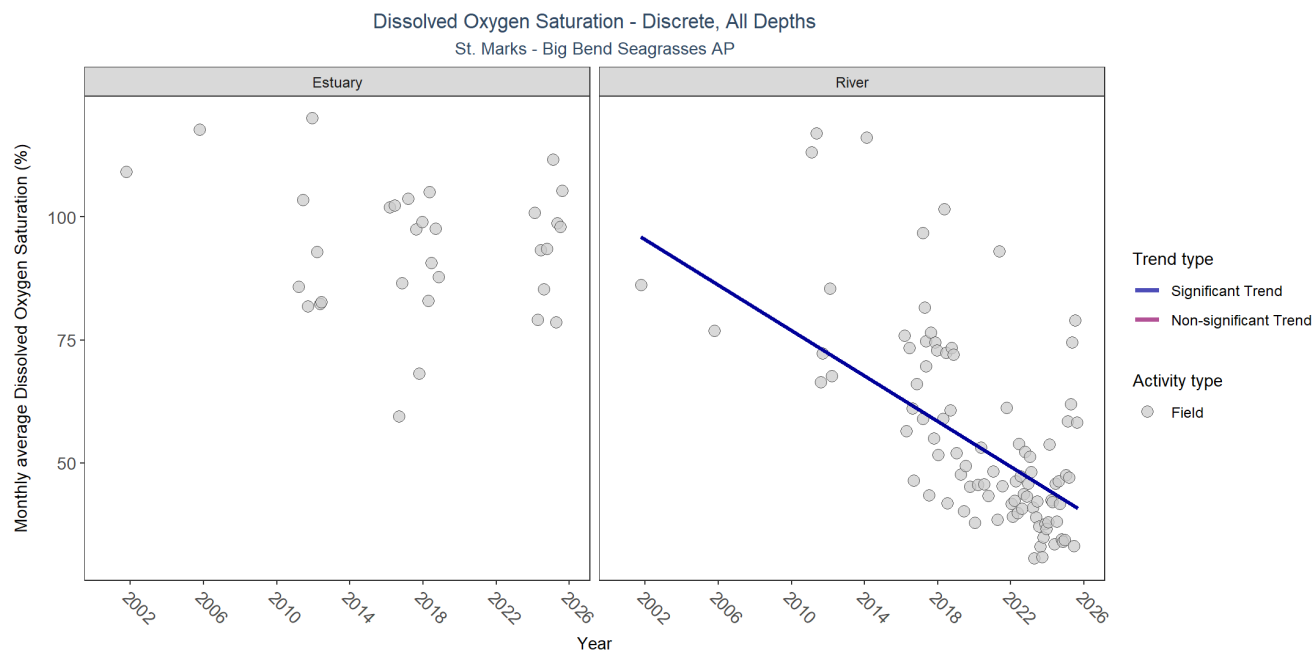
Dissolved Oxygen



$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

Dissolved Oxygen Saturation

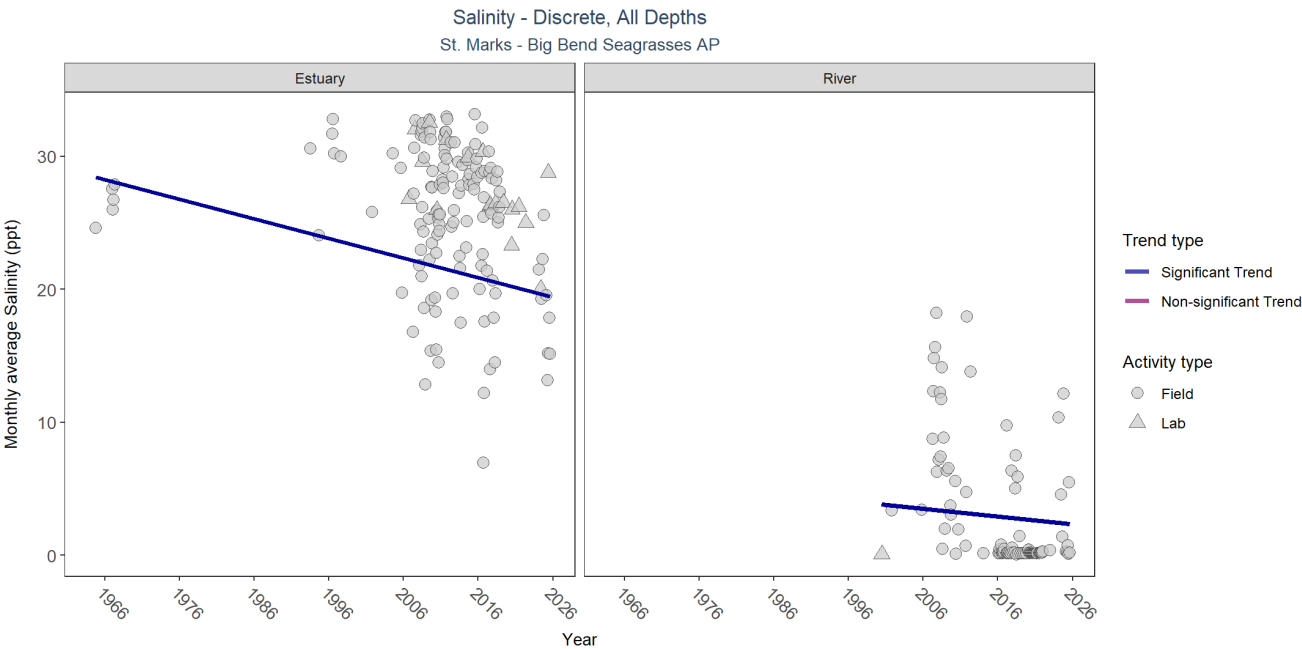


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	111	9	96.5	NA	NA	NA	NA	NA	NA	NA	NA
River	All	575	15	48.1	TRUE	-0.4568	0.0000	-2.306	97.68	7.65	0.7439	-1

$p < 0.00005$ appear as 0 due to rounding.

*SennIntercept is intercept value at beginning of
record for monitoring location*

Salinity

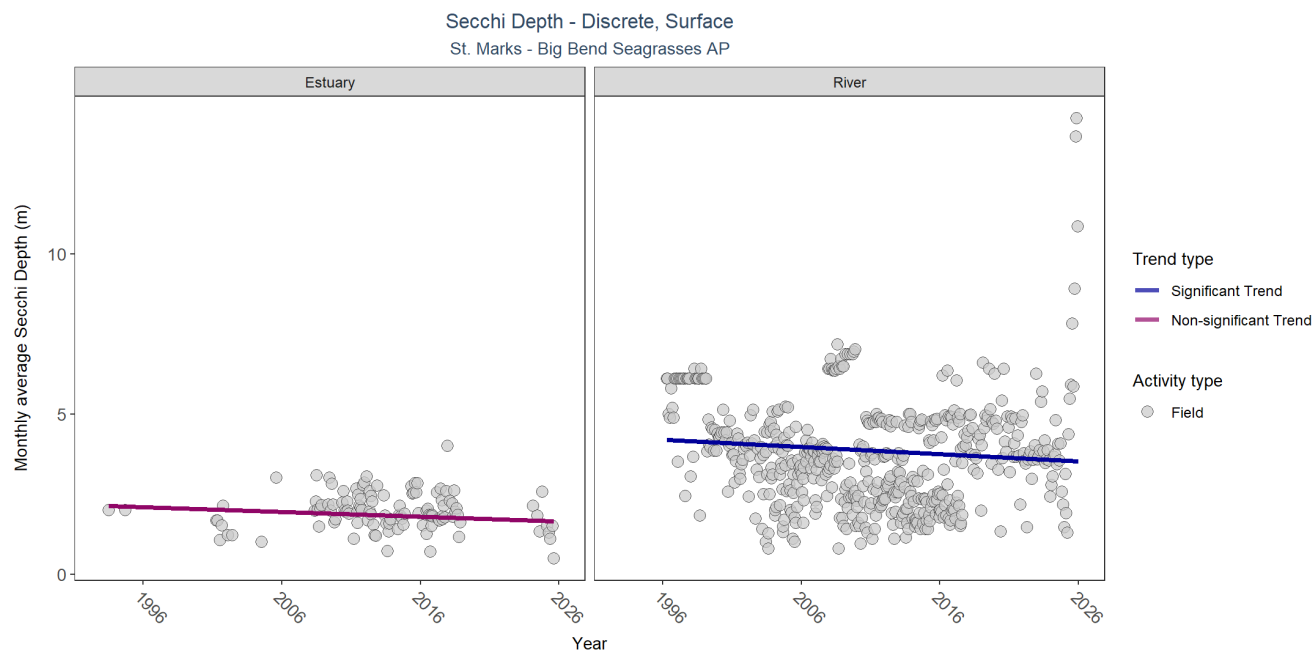


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	2852	28	27.8	TRUE	-0.1938	0.0052	-0.147	28.52	16.31	0.0911	-1
River	All	886	21	0.2	TRUE	-0.3005	0.0007	-0.058	3.87	8.25	0.6910	-2

$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of
record for monitoring location

Secchi Depth

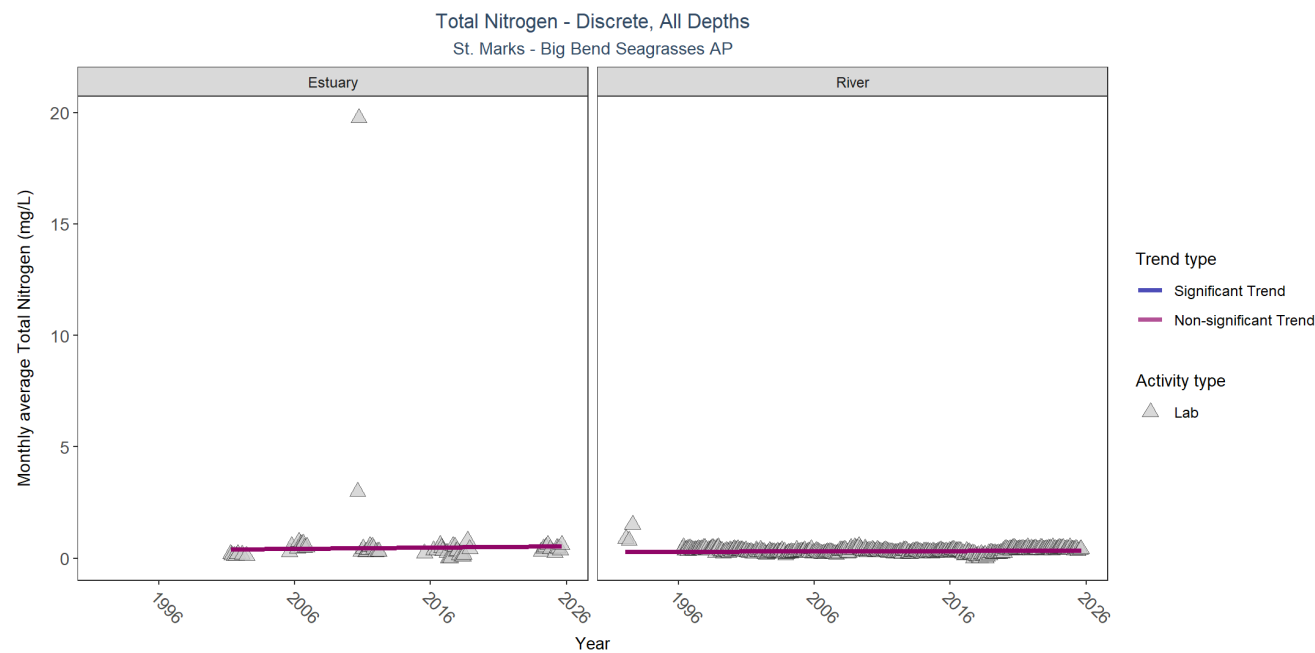


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	Surface	2071	19	2.000000	TRUE	-0.0622	0.1980	-0.015	2.15	10.33	0.4116	0
River	Surface	2551	30	3.048037	TRUE	-0.1194	0.0015	-0.023	4.21	6.12	0.8651	-1

$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

Total Nitrogen

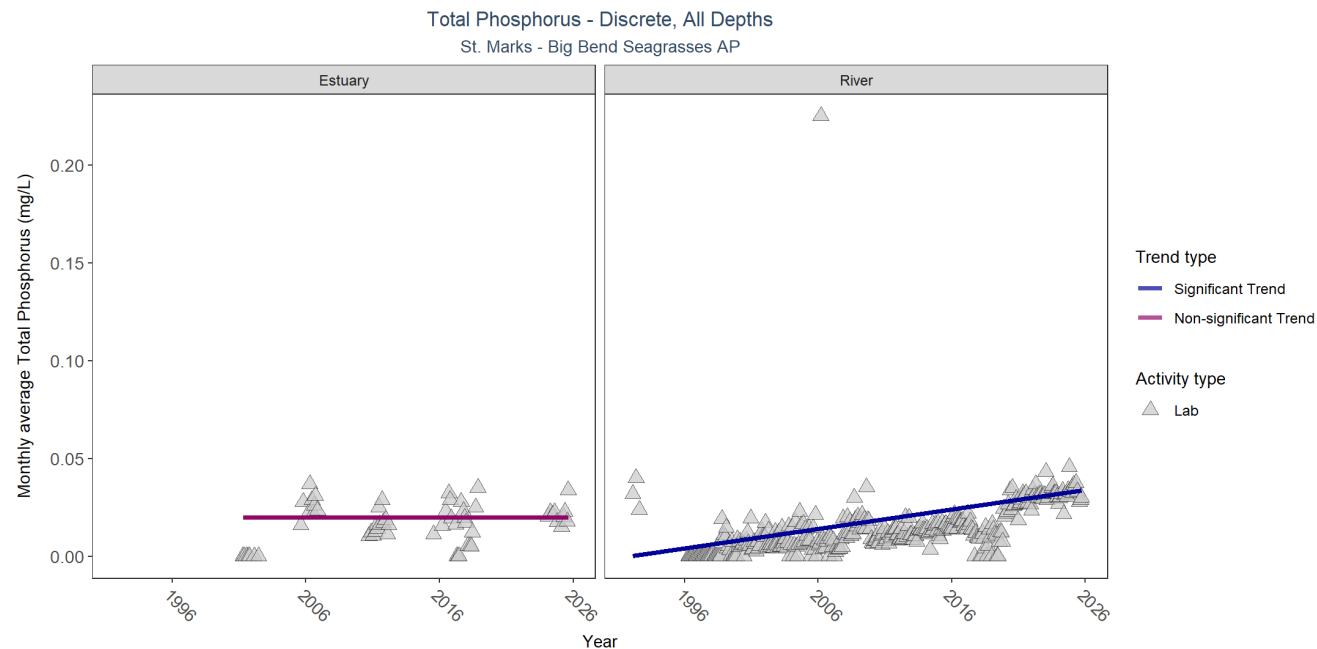


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	166	13	0.349	TRUE	0.1028	0.4935	0.006	0.39	4.60	0.9161	0
River	All	4251	31	0.330	TRUE	0.0346	0.3657	0.001	0.30	7.24	0.7794	0

p < 0.00005 appear as 0 due to rounding.

*SennIntercept is intercept value at beginning of
record for monitoring location*

Total Phosphorus

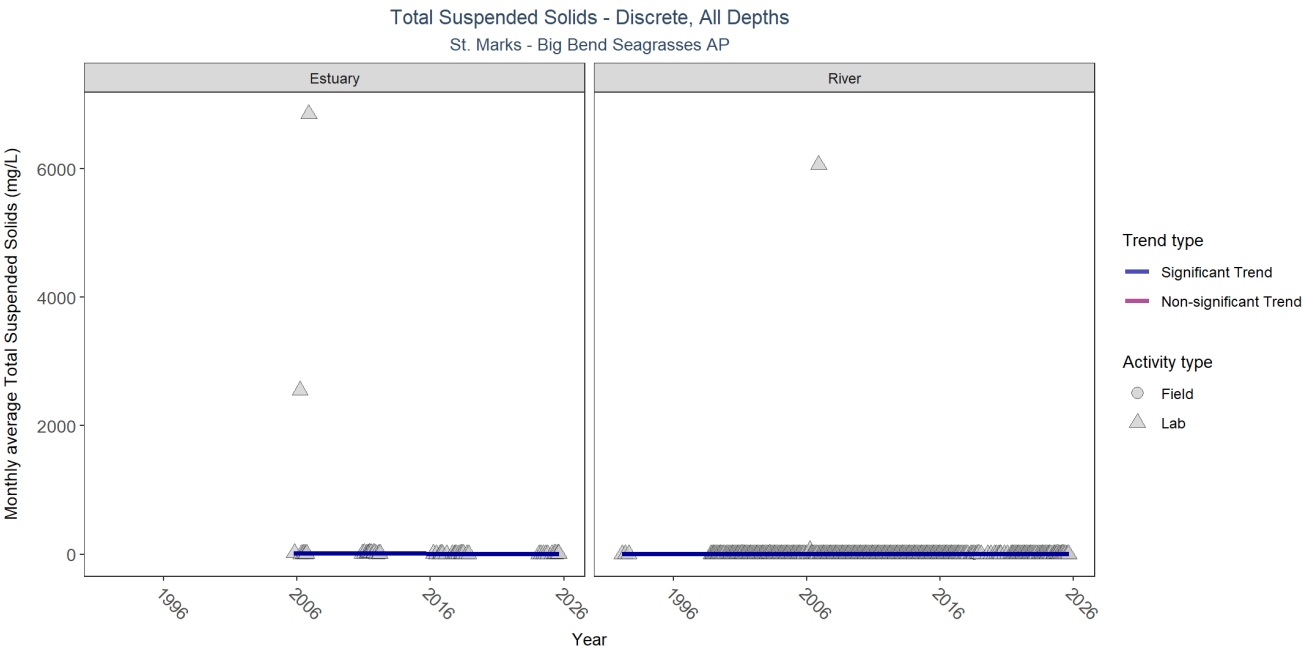


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	157	13	1.7e-02	TRUE	0.2276	0.0559	0.000	0.02	7.31	0.6963	0
River	All	2733	31	3.2e-05	TRUE	0.5542	0.0000	0.001	0.00	3.35	0.9852	2

$p < 0.00005$ appear as 0 due to rounding.

*SennIntercept is intercept value at beginning of
record for monitoring location*

Total Suspended Solids

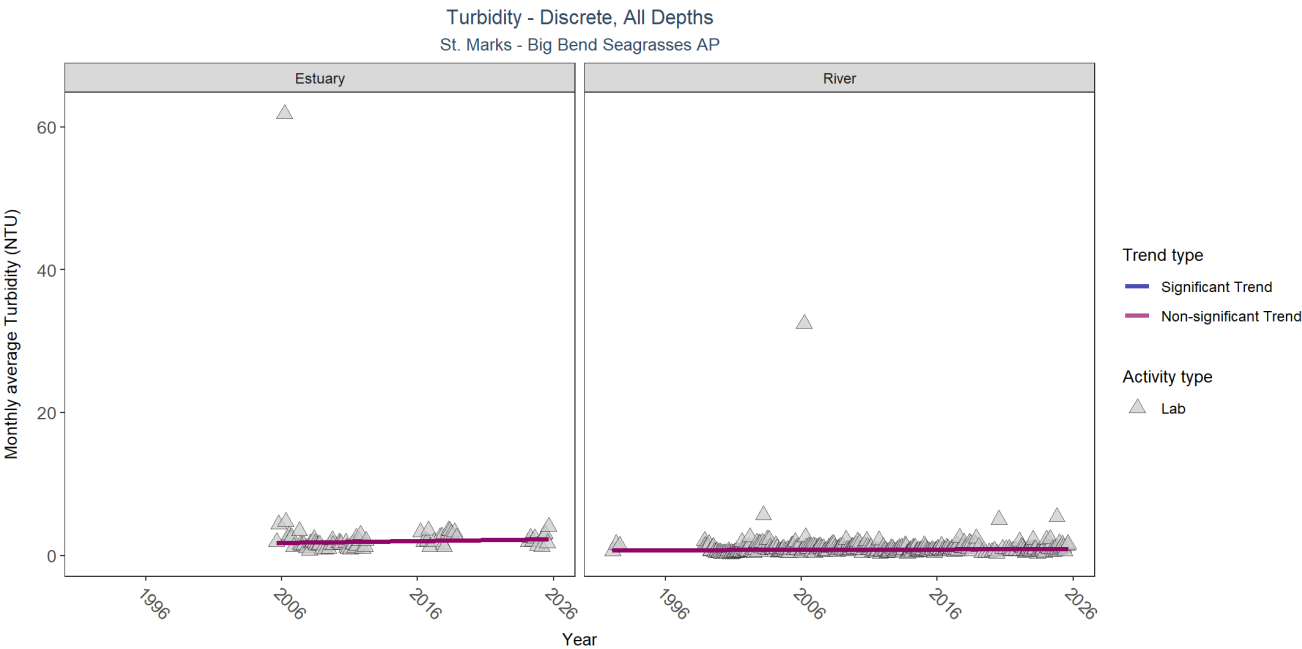


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	120	10	7	TRUE	-0.4519	0.0032	-0.375	10.63	14.91	0.1352	-1
River	All	831	29	3	TRUE	-0.4466	0.0000	-0.083	5.48	3.24	0.9872	-1

$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

Turbidity

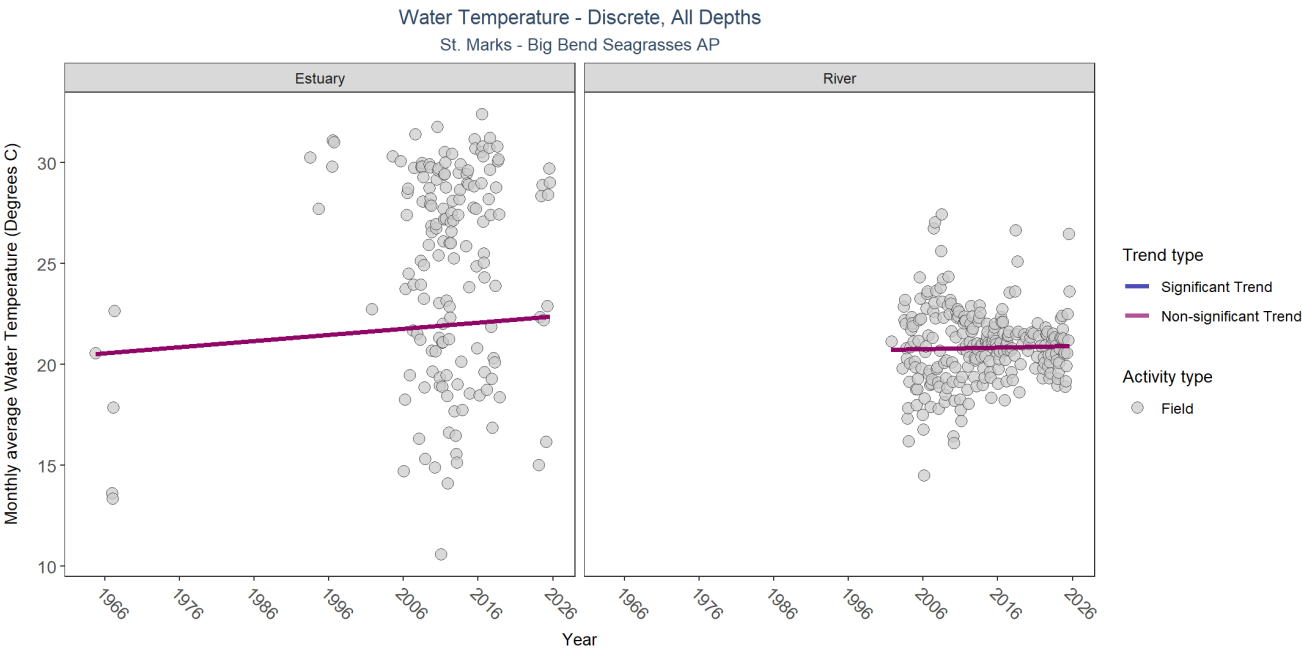


Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	478	13	1.60	TRUE	0.1909	0.1719	0.025	1.72	7.63	0.7460	0
River	All	846	29	0.75	TRUE	0.0686	0.0976	0.005	0.75	10.70	0.4687	0

$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

Water Temperature



Type	RelativeDepth	N_Data	N_Years	Median	Independent	tau	p	SennSlope	SennIntercept	ChiSquared	pChiSquared	Trend
Estuary	All	2770	23	27.4	TRUE	0.1101	0.1974	0.030	20.51	8.33	0.6835	0
River	All	1378	24	21.0	TRUE	0.0453	0.4680	0.007	20.73	48.46	0.0000	0

$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

pH



$p < 0.00005$ appear as 0 due to rounding.

SennIntercept is intercept value at beginning of record for monitoring location

Submerged Aquatic Vegetation

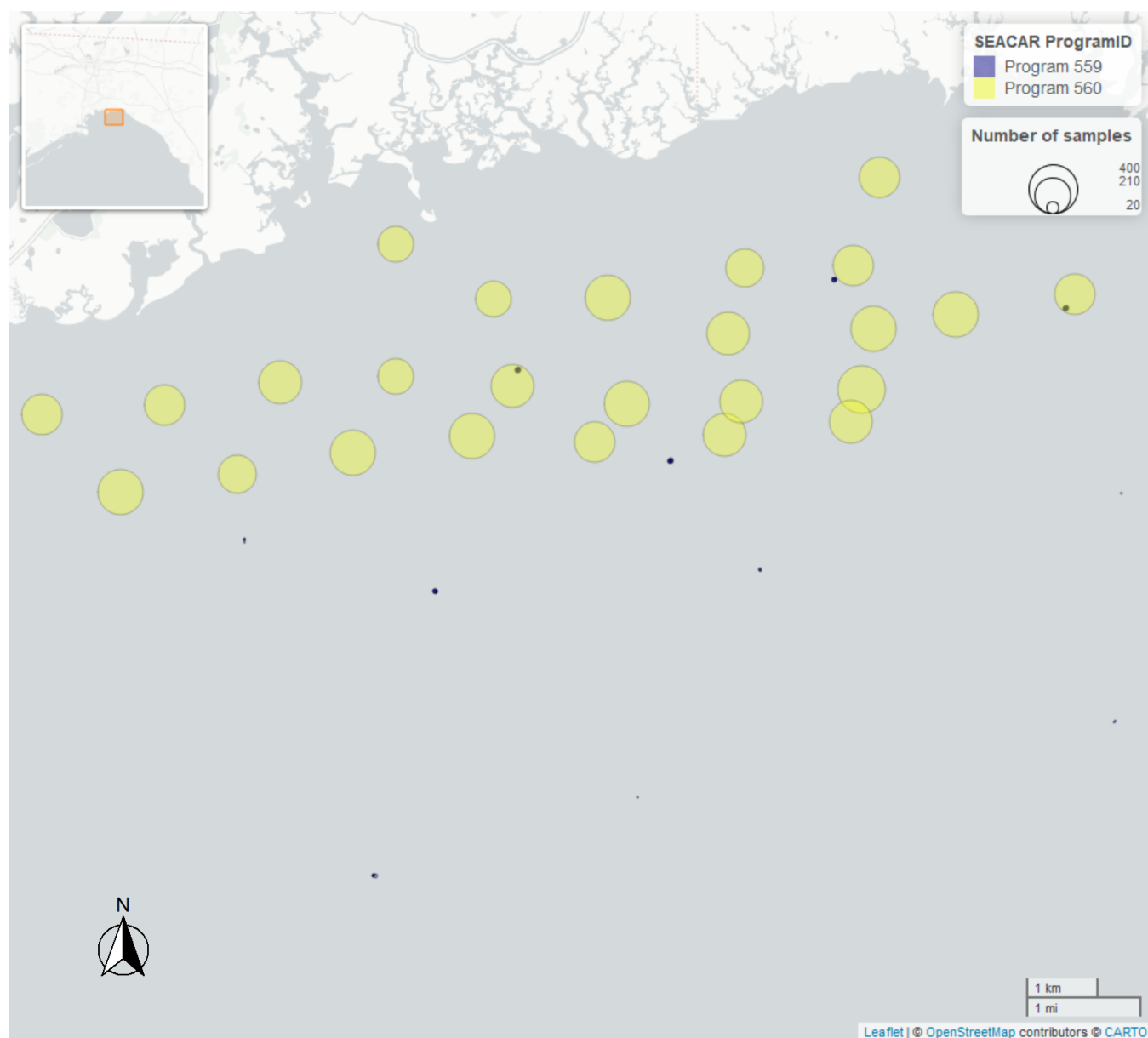


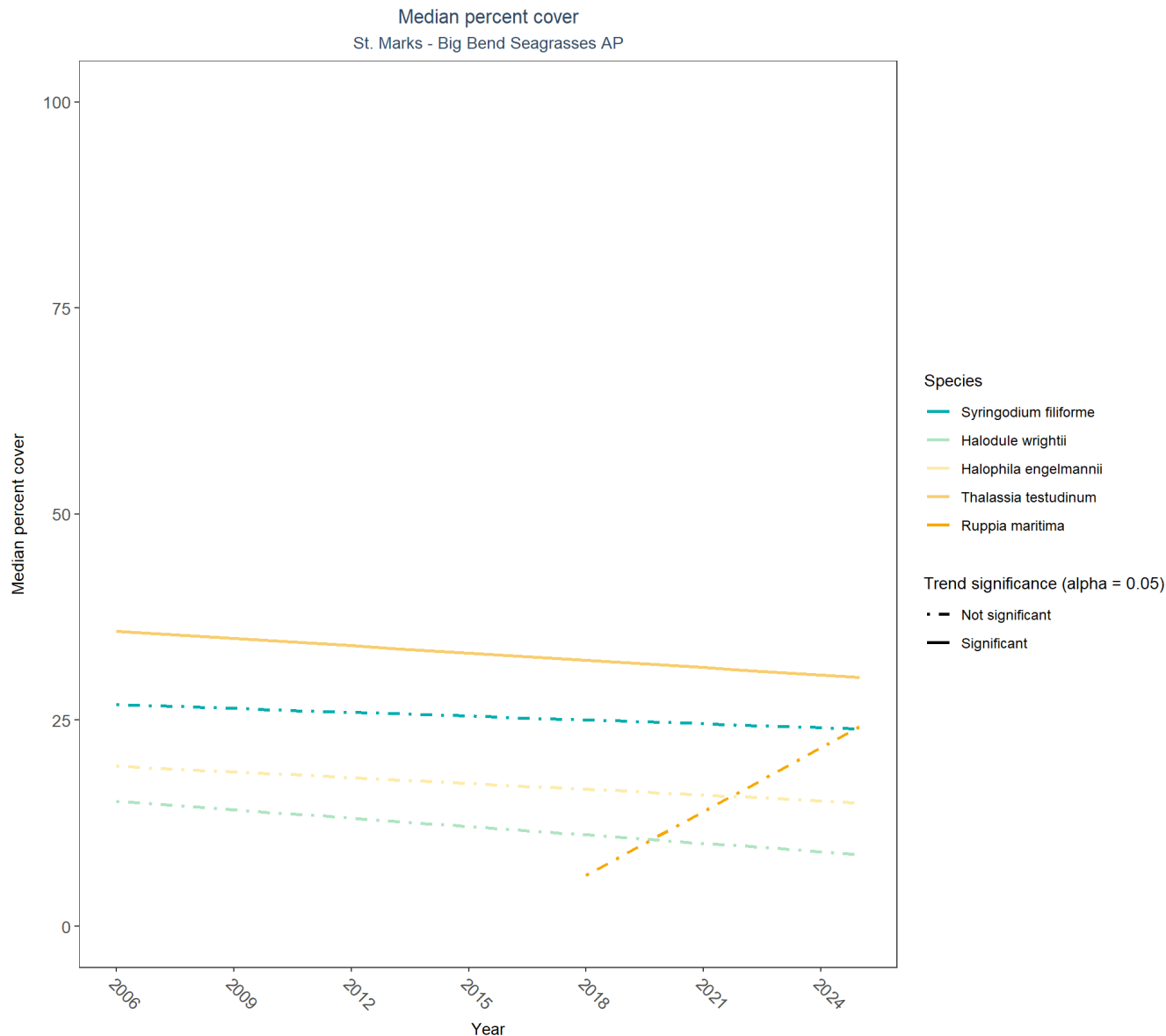
Table 3: Big Bend Seagrasses & Nature Coast Aquatic Preserves - Seagrass Monitoring - *Program 560*⁹

<i>N_Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>Collection Method</i>	<i>Sample Locations</i>
6326	2006	2025	Modified Braun Blanquet	25
1478	2022	2025	Percent Cover	25

Table 4: Northern Big Bend Seagrass Monitoring - *Program 559*¹⁰

<i>N_Data</i>	<i>YearMin</i>	<i>YearMax</i>	<i>Collection Method</i>	<i>Sample Locations</i>
131	2012	2018	Modified Braun Blanquet	54

Median Percent Cover - Species Trend Table



[1] "Table: Percent Cover Trend Analysis for St. Marks"

[2] " "

[3] "Species Trend Significance (0.05) Period of Record LME Intercept LME Slope p" [4] "-----"

19.9164 0.0409 0.8498" [5] "Drift algae No significant trend 2006 - 2025

17.1884 -0.3391 0.0619" [7] "Halophila engelmannii No significant trend 2006 - 2025

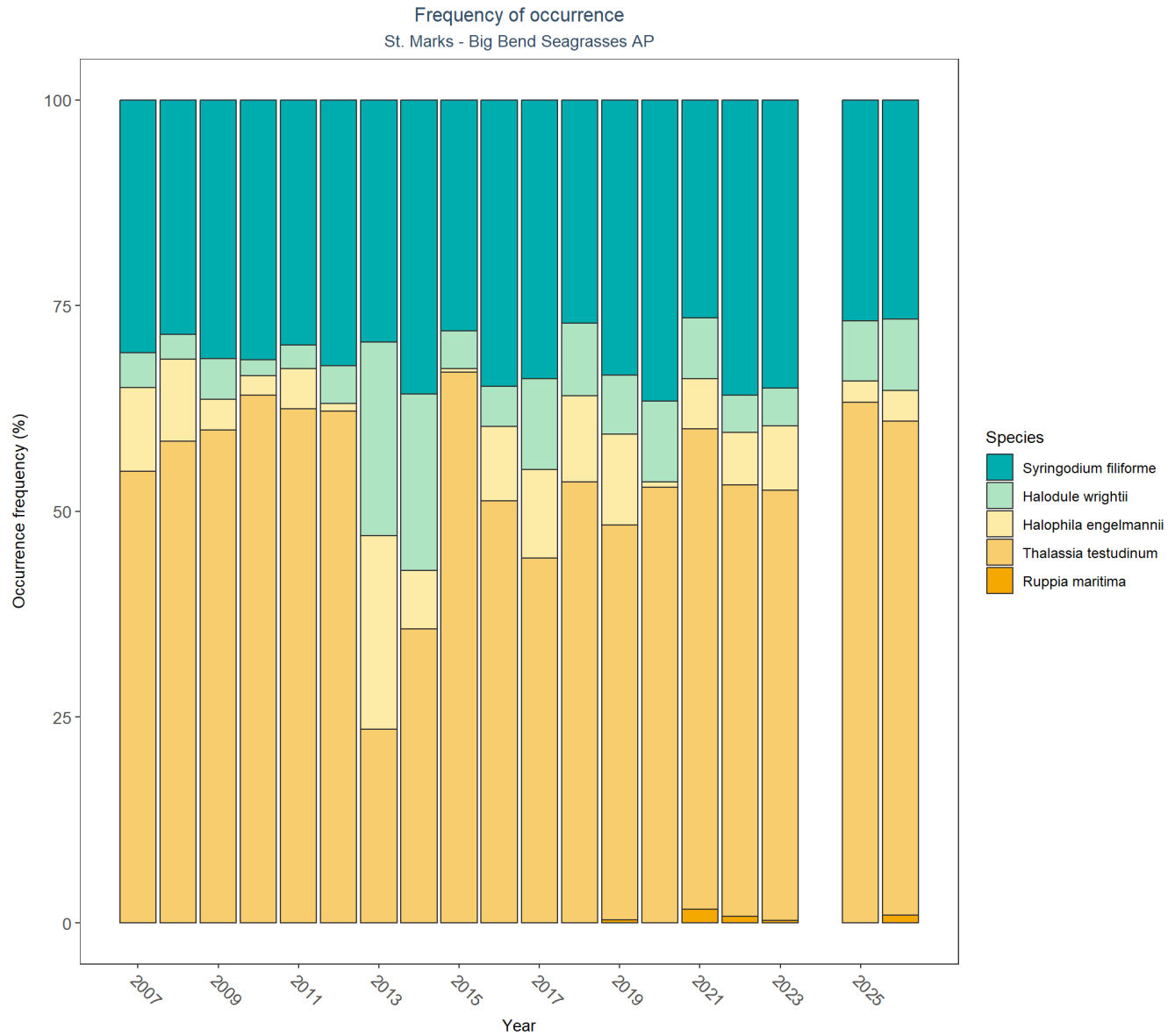
20.8083 -0.2320 0.3468" [8] "No grass in quadrat Insufficient data to calculate trend - - -"

[9] "Ruppia maritima No significant trend 2018 - 2025 -40.3540 2.5835 0.4946" [10] "Syringodium

filiforme No significant trend 2006 - 2025 27.8333 -0.1551 0.2791" [11] "Thalassia testudinum Significantly decreasing

trend 2006 - 2025 37.5753 -0.2948 0.0489" attr("format") [1] "simple" attr("class") [1] "knitr_kable"

Frequency of Occurrence



Submerged Aquatic Vegetation - Generalized Additive Models

Models generated for all systems

SEACAR Files used in the analysis

The following files were used in the analysis:

- *Combined_WQ_WC_NUT_Ammonia_Un-ionized_NH3-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Ammonium_NH4-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Fluorescent_Dissolved_Organic_Matter_FDOM-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Light_Extinction_Coefficient-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Nitrate_N-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Nitrite_N-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Nitrogen_inorganic-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Nitrogen_organic-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_NO2_3_Filtered-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_pH-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_PO4_Filtered-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Specific_Conductivity-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Total_Ammonia_N-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Total_Kjeldahl_Nitrogen_TKN-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-Jul-02.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-Jul-02.txt*

Threshold file used in QA process:

- [Database_Thresholds_20250409.xlsx](#)

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